

INTERNATIONAL AS PSYCHOLOGY

PS02

Unit 2 Biopsychology, Development and Research Methods 1

Mark scheme

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Version: 1.0 Final



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from www.oxfordaqa.com

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Section A: Biopsychology

Total for this section: 30 marks

Question	Marking Guidance	Total Marks
01	Using your knowledge of hemispheric lateralisation, identify the brain centres which are usually only located in the left hemisphere. Answer key: A – Language centres	1 AO1 = 1

Question	Marking Guidance	Total Marks
02	Describe what is meant by inhibition in the process of synaptic transmission. Award marks as follows: 3 marks: the description of inhibition in the process of synaptic transmission is detailed. The answer is clear with appropriate use of specialist terminology. 2 marks: the description of inhibition in the process of synaptic transmission lacks detail. The answer lacks clarity in places. 1 mark: the description of inhibition in the process of synaptic transmission is briefly presented. The answer is very limited/vague/muddled. Possible content • Inhibition is a decrease in the neural activity in the post-synaptic neuron. • It decreases the likelihood of an action potential being triggered. • Some neurotransmitters, eg serotonin, GABA and dopamine have an inhibitory effect making the receiving neuron less likely to fire. Credit other relevant material.	3 AO1 = 3

Question	Marking Guidance	Total Marks												
03	Noor is crossing a road, when suddenly a speeding car nearly hits her. Noor's breathing becomes fast and shallow and her heart beats quicker. Noor's legs are shaking. A few minutes later, Noor's breathing slows down again and she feels calmer. Use your knowledge of the autonomic nervous system to explain Noor's experiences. Possible content - Noor's experiences are due to the actions of the sympathetic and parasympathetic branches of the autonomic nervous system (ANS). - Due to the near accident, Noor's ANS activates the sympathetic branch which initiates the fight or flight response. - Adrenaline is released in Noor's body to bring on the physical changes she experiences. - Her breathing becomes shallow and fast to increase the oxygen supply to Noor's muscles which will enable her to respond to the approaching car. - Her heart beats quicker to pump the oxygenated blood rapidly to her muscles to fuel potential physical responses. - Noor's legs shake due to the increased energy supply to her muscles to enable her to respond quickly. - This is to mobilise energy for Noor's body to exert energy, eg jumping away from the car. - A few minutes later, the threat is over because Noor has not been hit by the car. The parasympathetic branch of Noor's ANS restores function to normal ('rest and digest'). - Noor's breathing slows down and she feels calmer because the body does not need to pump extra oxygen around the body. Credit other relevant material. <table border="1"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>3</td><td>Application of the knowledge of the autonomic nervous system to Noor's experiences is detailed and effective. The answer is clear with appropriate use of specialist terminology.</td><td>5–6</td></tr> <tr> <td>2</td><td>Application of the knowledge of the autonomic nervous system to Noor's experiences is relevant but detail/effectiveness is lacking. The answer lacks clarity in places. Specialist terminology is occasionally used appropriately.</td><td>3–4</td></tr> <tr> <td>1</td><td>Application of the knowledge of the autonomic nervous system to Noor's experiences is very limited.</td><td>1–2</td></tr> </tbody> </table>	Level	Description	Marks	3	Application of the knowledge of the autonomic nervous system to Noor's experiences is detailed and effective. The answer is clear with appropriate use of specialist terminology.	5–6	2	Application of the knowledge of the autonomic nervous system to Noor's experiences is relevant but detail/effectiveness is lacking. The answer lacks clarity in places. Specialist terminology is occasionally used appropriately.	3–4	1	Application of the knowledge of the autonomic nervous system to Noor's experiences is very limited.	1–2	6 AO2 = 6
Level	Description	Marks												
3	Application of the knowledge of the autonomic nervous system to Noor's experiences is detailed and effective. The answer is clear with appropriate use of specialist terminology.	5–6												
2	Application of the knowledge of the autonomic nervous system to Noor's experiences is relevant but detail/effectiveness is lacking. The answer lacks clarity in places. Specialist terminology is occasionally used appropriately.	3–4												
1	Application of the knowledge of the autonomic nervous system to Noor's experiences is very limited.	1–2												

		The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.		
	0	No creditable content.	0	

Question	Marking Guidance	Total Marks
04	Describe and evaluate research into plasticity and functional recovery after trauma. Possible content • Plasticity refers to the ability of the brain to change and adapt in the light of experiences, including trauma. • Description of studies investigating plasticity: eg Blakemore & Mitchell (1973) – development of visual cortex in cats demonstrating plasticity; Maguire (2000) – hippocampus changes in taxi drivers. • Functional recovery refers to the ability of the brain to reinstate lost function after brain damage. • Description of studies investigating functional recovery: eg Villablanca and Hovda (2000) – removal of one damaged hemisphere soon after birth; Kapar (1997) – doctors had better recovery after brain damage compared to the general population. • Explanation of mechanisms of recovery: eg reduction of swelling of brain tissue; axonal sprouting from surviving neurons; neurogenesis – growth of new neurons. • Reference to the role of age – more significant change possible in younger brains, eg Teuber (1975) found that soldiers under 20 had a much better recovery of movement and visual problems than older soldiers. Possible evaluation • Support from research studies, eg Maguire (2000), Villablanca and Hovda (2000). • Issues with generalisation of results from case studies or studies with small participant numbers. • Much of the research comes from non-human animals which leads to problems with extrapolation. • Knowledge about plasticity has given importance to rehabilitation and treatment in order to promote recovery. • Difficulty of predicting the level of recovery (low predictive validity). There are huge individual differences in recovery because many factors play a part, eg age, determination, practice, degree of damage. Credit other relevant material.	20 AO1 = 8 AO3 = 12

	<table> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> <tr> <td>4</td><td>Knowledge of research into plasticity and functional recovery is accurate and generally well detailed. Evaluation is mostly effective. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is used effectively.</td><td>15–20</td></tr> <tr> <td>3</td><td>Knowledge of research into plasticity and/or functional recovery is evident but there are occasional inaccuracies/omissions. There is some</td><td>11–14</td></tr> </table>	Level	Description	Marks	4	Knowledge of research into plasticity and functional recovery is accurate and generally well detailed. Evaluation is mostly effective. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is used effectively.	15–20	3	Knowledge of research into plasticity and/or functional recovery is evident but there are occasional inaccuracies/omissions. There is some	11–14	2 AO1 = 2
Level	Description	Marks									
4	Knowledge of research into plasticity and functional recovery is accurate and generally well detailed. Evaluation is mostly effective. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is used effectively.	15–20									
3	Knowledge of research into plasticity and/or functional recovery is evident but there are occasional inaccuracies/omissions. There is some	11–14									

		effective evaluation. The answer is clear and organised but occasionally lacks focus. Specialist terminology is used appropriately.		
	2	Limited knowledge of research into plasticity and/or functional recovery is present. Any evaluation is of limited effectiveness. The answer lacks clarity, accuracy and organisation in places. Specialist terminology is occasionally used appropriately.	6–10	
	1	Knowledge of research into plasticity and/or functional recovery is very limited. Evaluation is limited, poorly focused or absent. The answer as a whole lacks clarity, has many inaccuracies and is poorly organised. Specialist terminology is either absent or inappropriately used.	1–5	
	0	No creditable content.	0	

Section B: Cognitive Development

Total for this section: 30 marks

Question	Marking Guidance	Total Marks
05	What is meant by scaffolding in cognitive development? Award marks as follows: 2 marks: for a clear definition of scaffolding. 1 mark: for a muddled or vague definition of scaffolding. Possible content • Scaffolding is the help or support given to a person by a more knowledgeable other in order to complete a task. • Scaffolding consists of the activities provided by the more knowledgeable other to support the learner through the zone of proximal development. Credit alternative wording.	2 AO1 = 2

Question	Marking Guidance	Total Marks
06	Reginald is learning how to read. Using your knowledge of scaffolding, explain how Reginald’s father could help Reginald learn to read. Award marks as follows: 3 marks: the application of how scaffolding could be used to help Reginald learn to read is detailed. The answer is clear with appropriate use of specialist terminology. 2 marks: the application of how scaffolding could be used to help Reginald learn to read lacks detail. The answer lacks clarity in places. 1 mark: the application of how scaffolding could be used to help Reginald learn to read is briefly presented. The answer is very limited/vague/muddled. Possible content • The father might model reading to Reginald by reading aloud to him regularly. • He might provide explicit instructions to Reginald by sounding out letters and forming words. He should encourage Reginald to repeat this after him. • The father could point to words and help Reginald say them. They could then search for the same word on different pages to help Reginald recognise words. • Initially, the books Reginald reads with his father should have many pictures to help Reginald work out the meaning of the text he is reading. • When Reginald reads, the father should be encouraging and provide him with lots of praise. • With progress, the father can gradually reduce support as Reginald gains confidence and independence, eg Reginald might read aloud to his father and the books Reginald reads may have fewer pictures. Credit other relevant material.	3 AO2 = 3

Question	Marking Guidance	Total Marks
07	Describe the role of the mirror neuron system in social cognition. Award marks as follows: 3 marks: the description of the mirror neuron system is detailed. The answer is clear with appropriate use of specialist terminology. 2 marks: the description of the mirror neuron system lacks detail. The answer lacks clarity in places. 1 mark: the description of the mirror neuron system is briefly presented. The answer is very limited/vague/muddled. Possible content • Mirror neurons are a specific type of neuron in the brain that ‘fire’ both when an action is performed and when an action performed by someone else is observed. When we see someone carry out an action, eg drink water, our mirror neurons are activated as though we had drunk the water. • Mirror neurons play a role in several areas of social cognition, eg action, empathy, imitation, intention. • Observing someone in pain activates the same brain systems as if we were in pain – which is the foundation of empathy. • The mirror neuron system in humans is less activated in people low in empathy or people with ASD. Credit other relevant material.	3 AO1 = 3

Question	Marking Guidance	Total Marks
08	Eve watches Pete yawn and starts yawning herself. With reference to the mirror neuron system, briefly explain why Eve starts yawning. Award marks as follows: 2 marks: for a clear explanation of why Eve starts yawning using the mirror neuron system. 1 mark: for a muddled or vague explanation of why Eve starts yawning using the mirror neuron system. Possible application • As Eve watches Pete yawn, Eve's mirror neuron system will be activated, triggering a yawning response. This causes Eve to involuntarily mimic Pete's yawn. • Eve's response to Pete's yawn is a subconscious mirroring of his behaviour, a phenomenon commonly observed in social interactions. • Eve's yawn is not necessarily a sign of tiredness; instead, it reflects the neural mechanism of mirroring, indicating her brain's empathetic response to Pete's action. Credit other relevant material.	2 AO2 = 2

Question	Marking Guidance	Total Marks												
09	Describe the procedure of one study that investigated violation of expectation in very young children. Possible content Eg Drawbridge study • Familiarisation stage: the child sees a drawbridge swing 180 degrees. • Possible (expected) event: the child then views the possible event, where the motion of drawbridge is stopped by a block placed in the way. • Impossible (unexpected) event: the child views the impossible event, where the drawbridge swings all the way despite the block. • Measures taken: how long a child looks at the possible and impossible event. Credit the same elements for other relevant studies, eg tall/short carrot study; truck and screen study; Minnie Mouse and the window; tall/short rabbit study. <table border="1"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>2</td><td>Knowledge of the procedure of one study that investigated violation of expectation is accurate with some detail. The answer is clear with appropriate use of specialist terminology.</td><td>3–4</td></tr> <tr> <td>1</td><td>Knowledge of the procedure of one study that investigated violation of expectation is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.</td><td>1–2</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	2	Knowledge of the procedure of one study that investigated violation of expectation is accurate with some detail. The answer is clear with appropriate use of specialist terminology.	3–4	1	Knowledge of the procedure of one study that investigated violation of expectation is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.	1–2	0	No creditable content.	0	4 AO1 = 4
Level	Description	Marks												
2	Knowledge of the procedure of one study that investigated violation of expectation is accurate with some detail. The answer is clear with appropriate use of specialist terminology.	3–4												
1	Knowledge of the procedure of one study that investigated violation of expectation is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.	1–2												
0	No creditable content.	0												

Question	Marking Guidance	Total Marks															
10	Describe Piaget’s theory of cognitive development. Possible content • Refers to development of thinking skills which supposedly occurs in discrete stages, marked by qualitative differences. • This development is seen as a progressive reorganisation of mental processes as a result of biological maturation and environmental experiences. • Description of stages: Sensorimotor stage 0–2 years – development of object permanence; pre-operational stage 2–7 years – use of words as symbols for objects, egocentrism; concrete operational stage 7–11 years – beginning of operational or logical thought demonstrated by ability to work things out in head, conversation; formal operational stage 11+ years – ability to abstract thought and reasoning, logical thinking, hypothesis testing. • Description of other key components/processes: schema and adaptation (assimilation, accommodation, equilibration). Credit other relevant material. <table border="1"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>3</td><td>Knowledge of Piaget’s theory of cognitive development is detailed and appropriate. The answer is clear with appropriate use of specialist terminology.</td><td>5–6</td></tr> <tr> <td>2</td><td>Knowledge of Piaget’s theory of cognitive development is relevant but detail is lacking. The answer lacks clarity in places. Specialist terminology is occasionally used appropriately.</td><td>3–4</td></tr> <tr> <td>1</td><td>Knowledge of Piaget’s theory of cognitive development is very limited. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.</td><td>1–2</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	3	Knowledge of Piaget’s theory of cognitive development is detailed and appropriate. The answer is clear with appropriate use of specialist terminology.	5–6	2	Knowledge of Piaget’s theory of cognitive development is relevant but detail is lacking. The answer lacks clarity in places. Specialist terminology is occasionally used appropriately.	3–4	1	Knowledge of Piaget’s theory of cognitive development is very limited. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.	1–2	0	No creditable content.	0	6 AO1 = 6
Level	Description	Marks															
3	Knowledge of Piaget’s theory of cognitive development is detailed and appropriate. The answer is clear with appropriate use of specialist terminology.	5–6															
2	Knowledge of Piaget’s theory of cognitive development is relevant but detail is lacking. The answer lacks clarity in places. Specialist terminology is occasionally used appropriately.	3–4															
1	Knowledge of Piaget’s theory of cognitive development is very limited. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.	1–2															
0	No creditable content.	0															

Question	Marking Guidance	Total Marks
11	Explain one strength of Piaget’s theory of cognitive development. Award marks as follows: 3 marks: the strength of Piaget’s theory of cognitive development is detailed. The answer is clear with appropriate use of specialist terminology. 2 marks: the strength of Piaget’s theory of cognitive development lacks detail. The answer lacks clarity in places. 1 mark: the strength of Piaget’s theory of cognitive development is briefly presented. The answer is very limited/vague/muddled. Possible content • Piaget used a variety of methods, including carefully controlled observations and devising tasks for children to complete. These have been replicated by other developmental psychologists who confirmed some of Piaget’s findings (establishing reliability). • Supporting research evidence: eg Piaget and Inhelder (1967) – three mountains task. • Influential research with historical importance which has led to real-life applications – implementations in educational policies and teaching practice by regarding children as active learners who have to be ‘ready’ for certain learning. Credit other relevant material.	3 AO3 = 3

Question	Marking Guidance	Total Marks												
12	Describe one of the Sally–Anne studies investigating theory of mind. Possible content • Description of a/one Sally–Anne study (Baron-Cohen, Leslie & Frith, 1985). • The child participant is shown two dolls, Sally and Anne. Sally has a basket, and Anne has a box. • The researcher explains Sally places her marble in the basket and leaves the room. While Sally is gone, Anne moves the marble from the basket to the box. • The researcher then asks the child three questions; “Where is the marble really?”; “Where was the marble in the beginning?” and “Where will Sally look for her marble?” (the critical question). • The child’s response is analysed to determine whether they understand that Sally, who has not witnessed Anne moving the marble, will mistakenly think the marble is still in the basket. • Description of other research using false belief task, eg Maxi test by Wimmer and Perner, (1983) and/or appearance reality task, eg smarties task by Perner, Leekam & Wimmer, (1987). Credit other relevant material. <table border="1"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>2</td><td>Knowledge of one of the Sally-Anne studies is accurate with some detail. The answer is clear with appropriate use of specialist terminology.</td><td>3–4</td></tr> <tr> <td>1</td><td>Knowledge of one of the Sally-Anne studies is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.</td><td>1–2</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	2	Knowledge of one of the Sally-Anne studies is accurate with some detail. The answer is clear with appropriate use of specialist terminology.	3–4	1	Knowledge of one of the Sally-Anne studies is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.	1–2	0	No creditable content.	0	4 AO1 = 4
Level	Description	Marks												
2	Knowledge of one of the Sally-Anne studies is accurate with some detail. The answer is clear with appropriate use of specialist terminology.	3–4												
1	Knowledge of one of the Sally-Anne studies is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.	1–2												
0	No creditable content.	0												

Question	Marking Guidance	Total Marks
13	Explain one limitation of the Sally–Anne studies. Award marks as follows: 3 marks: the explanation of one limitation of Sally-Anne studies is detailed. The answer is clear with appropriate use of specialist terminology. 2 marks: the explanation of one limitation of the Sally-Anne studies lacks detail. The answer lacks clarity in places. 1 mark: the explanation of one limitation of the Sally-Anne studies is briefly presented. The answer is very limited/vague/muddled. Possible limitation • Requires participants to have sufficiently developed language skills. Therefore not suitable for very young children or children with ASD. • Generalisation issues using a sample of children with developmental deficits, eg children with ASD. • Individual differences/nurture – children from larger and/or more affluent families outperform their peers suggesting an environmental influence on theory of mind development. Credit other relevant material.	3 AO3 = 3

Section C: Research Methods 1

Total for this section: 30 marks

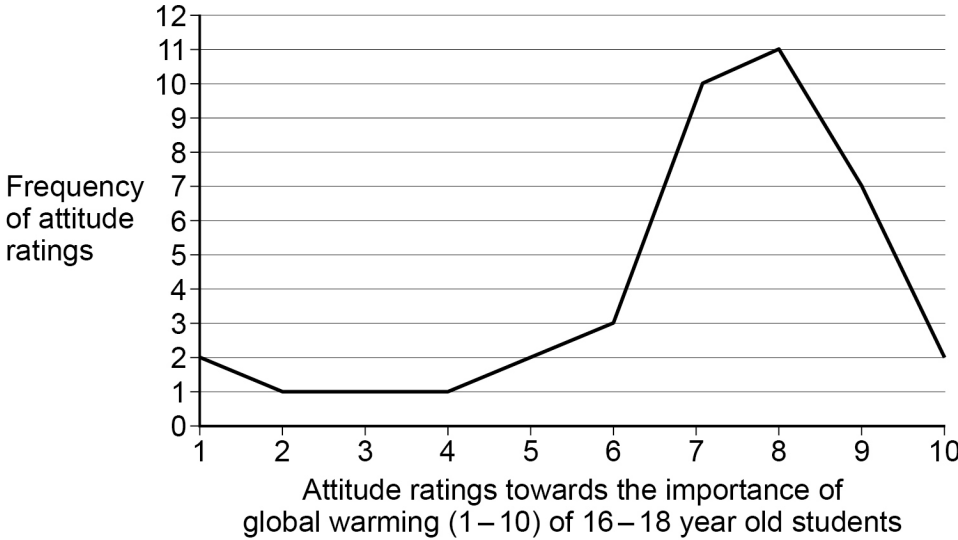
	Researchers wanted to investigate the attitudes of 16–18-year-old students towards environmental issues. They created a questionnaire that contained both open questions and closed questions. The researchers carried out a pilot study of the questionnaire. After the pilot study, they selected 40 students aged 16–18 from a local school to take part in their investigation.	
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Question	Marking Guidance	Total Marks
14	Briefly explain one reason why the researchers carried out a pilot study of the questionnaire. Award marks as follows: 2 marks: for a clear explanation of one reason why the researchers carried out a pilot study of the questionnaire. 1 mark: for a muddled or vague explanation of one reason why the researchers carried out a pilot study with limited application. Possible content • The pilot study would enable the researchers to check that the content of the questions concerns the attitudes of students towards environmental issues. • The researchers could check whether the students feel comfortable answering all questions because if they do not feel comfortable, this would lead to ethical issues.	2 AO2 = 2

Question	Marking Guidance	Total Marks
15	Identify the population and the sample in this investigation. Award marks as follows: Award 1 mark for population: 16–18-year-old students. Award 1 mark for sample: (40, 16–18-year-old) students from a local school.	2 AO2 = 2

Question	Marking Guidance	Total Marks
16	Write an open question the researchers could have used in their questionnaire. Award marks as follows 1 mark: for a suitable open question, eg “Explain what you can do to protect the environment.” 0 marks: for a closed question.	1 AO2 = 1

Question	Marking Guidance	Total Marks																						
17	One of the questions asked the 16–18-year-old students to rate their attitudes towards global warming on a scale of 1–10, where 1 meant not very important and 10 meant most important. The results are shown in Table 1 below. Table 1 Frequency of rating scores on the attitudes scale (1–10). <table><tr><th>Rating score of attitude scale (1–10)</th><th>Frequency</th></tr><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>1</td></tr><tr><td>3</td><td>1</td></tr><tr><td>4</td><td>1</td></tr><tr><td>5</td><td>2</td></tr><tr><td>6</td><td>3</td></tr><tr><td>7</td><td>10</td></tr><tr><td>8</td><td>11</td></tr><tr><td>9</td><td>7</td></tr><tr><td>10</td><td>2</td></tr></table>	Rating score of attitude scale (1–10)	Frequency	1	2	2	1	3	1	4	1	5	2	6	3	7	10	8	11	9	7	10	2	
Rating score of attitude scale (1–10)	Frequency																							
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Question	Marking Guidance	Total Marks																						
17.1	Using the data from Table 1, draw a line graph. Provide a title for your graph, and fully label both axes. Award marks as follows: 1 mark for a title with fully operationalised variables, eg (line graph showing) the frequency of attitude ratings (1-10) towards global warming. 1 mark for correctly labelling the Y axis; Frequency (of attitude ratings). 1 mark for correctly labelling the X axis; Attitude ratings of 16–18-year-old students towards the importance of global warming (1-10). 1 mark for correctly plotting the data. Line graph showing the frequency of attitude ratings (1 – 10) towards global warming  <table data-bbox="303 799 1265 1337"><caption>Data points for the line graph</caption><thead><tr><th>Attitude Rating (X-axis)</th><th>Frequency of Attitude Ratings (Y-axis)</th></tr></thead><tbody><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>1</td></tr><tr><td>3</td><td>1</td></tr><tr><td>4</td><td>1</td></tr><tr><td>5</td><td>2</td></tr><tr><td>6</td><td>3</td></tr><tr><td>7</td><td>10</td></tr><tr><td>8</td><td>11</td></tr><tr><td>9</td><td>7</td></tr><tr><td>10</td><td>2</td></tr></tbody></table>	Attitude Rating (X-axis)	Frequency of Attitude Ratings (Y-axis)	1	2	2	1	3	1	4	1	5	2	6	3	7	10	8	11	9	7	10	2	4 AO2 = 4
Attitude Rating (X-axis)	Frequency of Attitude Ratings (Y-axis)																							
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7	10																							
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9	7																							
10	2																							

Question	Marking Guidance	Total Marks
17.2	Using the data from Table 1, calculate the mean, median and mode. Show your workings. Award marks as follows For mean: 2 marks for the correct answer: $\text{mean} = 280/40 = 7$ 1 mark if the answer is incorrect but there are some appropriate calculations. For median: 2 marks for the correct answer: $\text{median} = 7+8/2 = 7.5$ 1 mark if the answer is incorrect but there are some appropriate calculations. For mode: 2 marks for the correct answer: most frequent rating is 8 with 11 ratings; mode = 8 1 mark if the answer is incorrect but there are some appropriate calculations.	6 AO2 = 6

Question	Marking Guidance	Total Marks
17.3	Using the mean, median and mode you calculated in your answer to Question 17.2, what do these statistics tell us about the distribution of the frequency ratings? Award 1 mark for each of the following bullet points (maximum of 4 marks): • it is a negative skewed/not a normal distribution • the bulk of the data is at the higher end of the scale; most scores are above/bigger than the mean 7 • the median 7.5 is larger than the mean 7 • the mode 8 is larger than the median 7.5 • the distribution is not symmetrical as mean, mode and median are all different • the statistics indicate that the majority/most of the students think that global warming is an important issue. Credit other relevant material.	4 AO2 = 4

	To gain a better understanding of attitudes towards environmental issues, the researchers carried out an unstructured interview with some of the 16–18-year-old students.	
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Question	Marking Guidance	Total Marks
18	What is meant by an unstructured interview? Award marks as follows: 2 marks: for a clear definition of an unstructured interview. 1 mark: for a muddled or vague definition of an unstructured interview. Possible content • An unstructured interview is a type of interview which is guided by themes rather than predetermined questions. • An unstructured interview enables the interviewer to respond to the answers given by the participant, eg to seek clarification or pursue a new line of enquiry. Credit alternative wording.	2 AO1 = 2

Question	Marking Guidance	Total Marks
19.1	Explain why having the right to withdraw is important in the design and conduct of psychological studies. Award 3 marks as follows: 3 marks: the explanation of why having the right to withdraw is important is detailed. The answer is clear with appropriate use of specialist terminology. 2 marks: the explanation of why having the right to withdraw is important lacks detail. The answer lacks clarity in places. 1 mark: the explanation of why having the right to withdraw is important is briefly presented. The answer is very limited/vague/muddled. Possible content • Provides an added layer of protection from harm for the participants who may experience emotional discomfort or stress. • It ensures that the participants have control over their involvement in the study. • When participants are aware that they can leave the study at any time, they are more likely to be honest and forthcoming in their responses. This enhances the validity and reliability of the data collected. • The ability for the participants to withdraw their data at the end of the study is important to reduce any possible distress or harm. Credit other relevant material. Note: A definition of the right to withdraw without any reference to why it is important will not be credited.	3 AO1 = 3

Question	Marking Guidance	Total Marks
19.2	Explain how the researchers could have ensured that the 16–18-year-old students knew that they had the right to withdraw from the interview. Award 3 marks as follows: 3 marks: the explanation of how the researchers could have ensured that the 16–18-year-old students knew that they had the right to withdraw is detailed. The answer is clear with appropriate use of specialist terminology. 2 marks: the explanation of how the researchers could have ensured that the 16–18-year-old students knew that they had the right to withdraw lacks detail. The answer lacks clarity in places. 1 mark: the explanation of how the researchers could have ensured that the 16–18-year-old students knew that they had the right to withdraw is briefly presented. The answer is very limited/vague/muddled. Possible content • The researchers should have explained to the 16-18-year-old students as part of the consent process that completing the interview is voluntary and that they can withdraw at any point, without needing to provide a reason or facing any negative consequences.	3 AO2 = 3

	- It is the researchers' responsibility to remind the 16-18-year-old students of their right to withdraw if they appear distressed while answering the questions in the interview. - The 16-18-year-old students should be told as part of the debrief how they can get in touch with the researchers if they want to withdraw their interview responses. Credit other relevant material.	
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Question	Marking Guidance	Total Marks
20	At the end of each interview, the researchers conducted a debrief with the student. Explain what information the researchers should have included in the debrief. Award marks as follows: 3 marks: the explanation of the information provided in the debrief is detailed. The answer is clear with appropriate use of specialist terminology. 2 marks: the explanation of the information provided in the debrief lacks detail. The answer lacks clarity in places. 1 mark: the explanation of the information provided in the debrief is briefly presented. The answer is very limited/vague/muddled. Possible content - The researchers should have reminded the 16-18-year-old students of the aim of the interview which was to investigate attitudes towards environmental issues. - The researchers should have reminded the 16-18-year-old students that they can withdraw their answers to the interview if they want to. - The researchers should have reassured the 16-18-year-old students that the interview was confidential and that their privacy was guaranteed. - The researchers should offer access to help during the debrief in case the 16-18-year-old students felt that the interview brought up anything they wished to discuss further. Credit other relevant material.	3 AO2 = 3

PS02 grid

	AO1	AO2	AO3	Total
Section A				
01	1			1
02	3			3
03		6		6
04	8		12	20
Section B				
05	2			2
06		3		3
07	3			3
08		2		2
09	4			4
10	6			6
11			3	3
12	4			4
13			3	3
Section C				
14		2		2
15		2		2
16		1		1
17.1		4		4
17.2		6		6
17.3		4		4
18	2			2
19.1	3			3
19.2		3		3
20		3		3
Unit total	36	36	18	90